What Is UML?

Drawing software before you build it

How do you draw a design?

UML

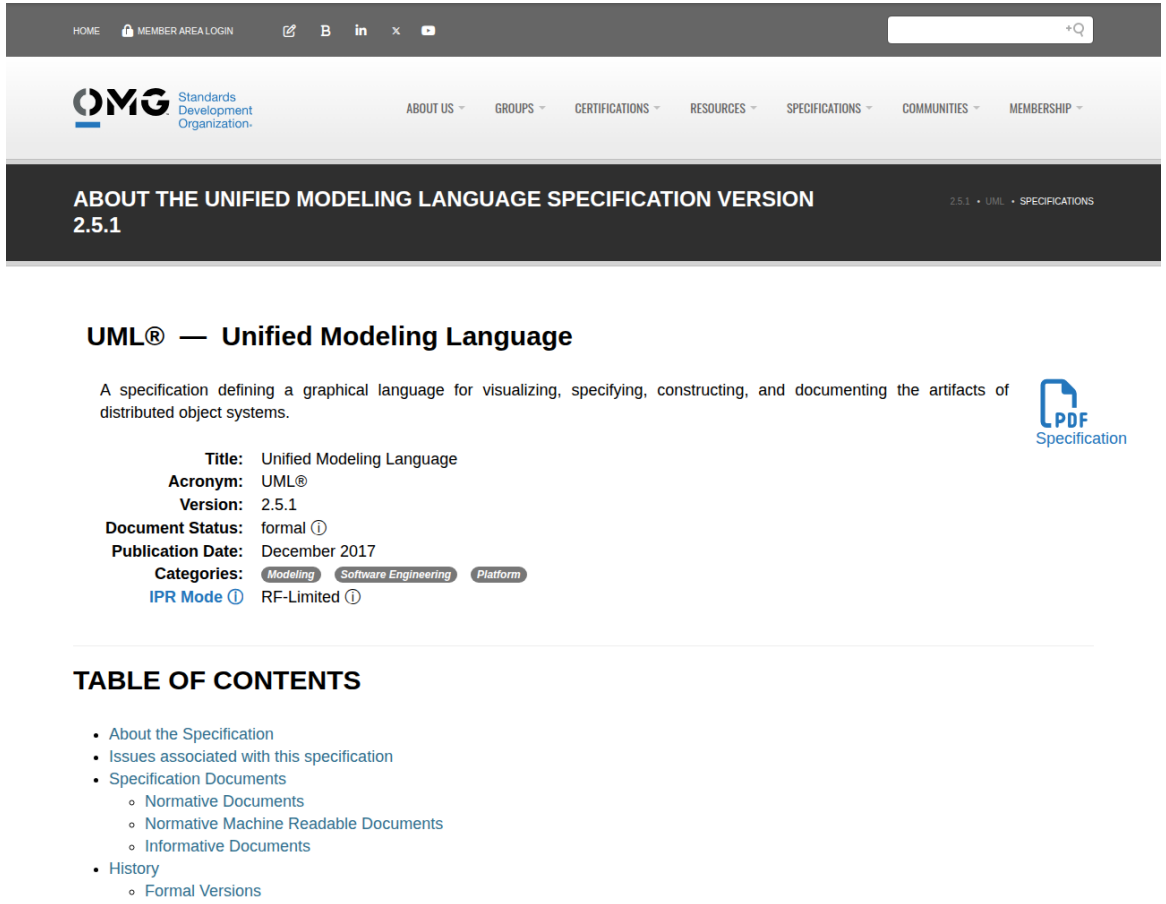
A standard visual language for software designs

The Official Standard

Maintained by the OMG

Current version 2.5.1

Published December 2017



Source

Where It Came From

1995

Merged

Three design methods combine

1997

OMG

Adopted as a standard

2005

ISO

Becomes an ISO standard

2017

UML 2.5.1

The version in use today

Source

Fourteen Diagrams, Two Families

Structure: what it is	Behaviour: what it does
Class	Use case
Component	Sequence
Deployment	State machine

Class diagram

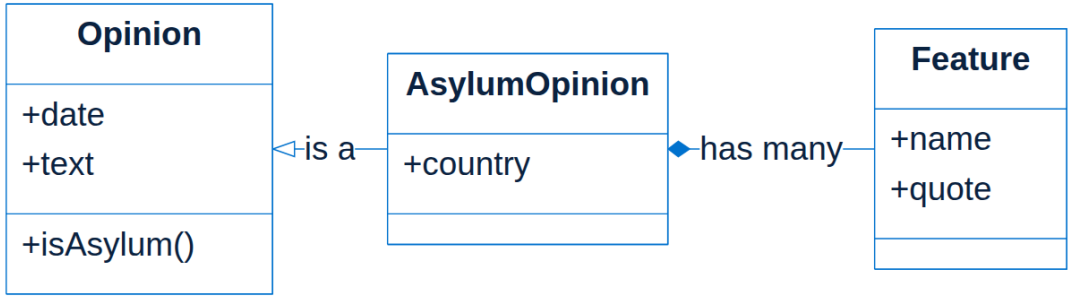
The things in a system and how they relate

A Class Diagram, Concretely

Each box is a class

Each line is a link

Modelling ninth_circuit



Source

Reading a Class Box

Part of the box	What it holds
Top	The class name
Middle	Its attributes
Bottom	Its operations

The Lines Between Classes

Line end	Means
Hollow triangle	Is a kind of
Filled diamond	Owns, and dies with
Hollow diamond	Has, but shares
Plain line	Knows about

Sequence diagram

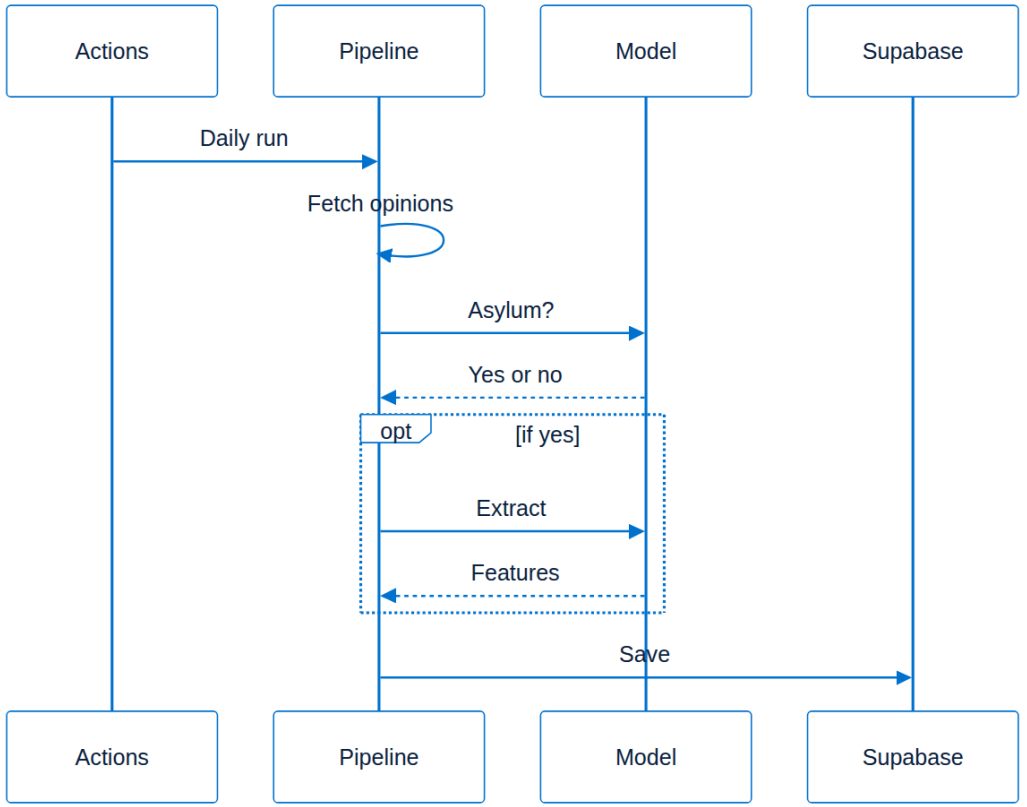
Who talks to whom, and in what order

A Sequence Diagram, Concretely

Time runs downward

Solid arrows: requests

Dashed arrows: replies



Source

Boxes That Change the Flow

opt: happens only if true

alt: one branch or the other

loop: repeats while true

par: happens at the same time

State machine

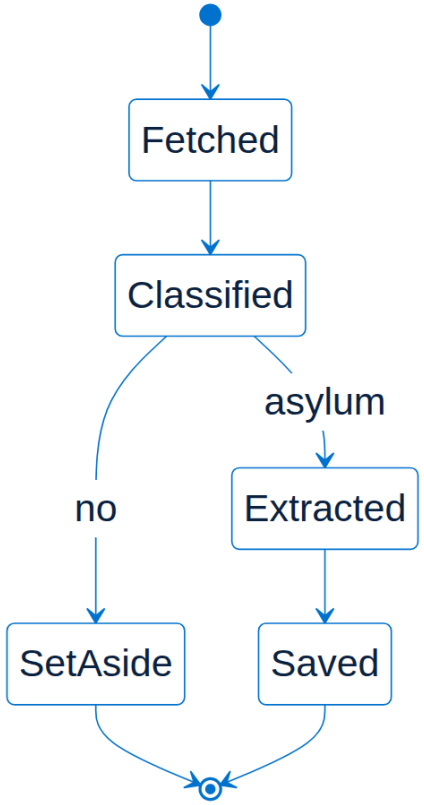
The stages one thing moves through

A State Diagram, Concretely

Filled dot: the start

Ringed dot: the end

Labels say why it moves



Source

Which Diagram, When

If you are asking	Draw a
What is it made of?	Class diagram
Who talks to whom?	Sequence diagram
What stages can it be in?	State diagram
What can a user do?	Use case diagram

Tools for Drawing It

Pen and paper, honestly

Mermaid: diagrams written as text

PlantUML: also text

draw.io: drag and drop

Good Habits

Draw to explain, not to archive

One question per diagram

Leave out what does not matter

Keep it next to the code

Sources

Slide	Taken from
Specification	omg.org/spec/UML
History, diagram types	Wikipedia, UML
Example diagrams	Drawn with Mermaid
What they model	ninth_circuit README

What would you draw first?